Eduardo Sampaio Pimenta

Mathematician — Probability Theory and Stochastic Processes

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Research interests

Probability theory, stochastic processes, scaling limits, invariance principles, boundary random walks, Feller processes, dynamic percolation and evolving random environments.

Academic experience and education

- **Postdoctoral research**, Federal University of Minas Gerais (UFMG), 2026–present.
- **Postdoctoral research**, University of Augsburg, 2025–2026.
- **PhD in Mathematics**, Federal University of Bahia (UFBA), 2021–2025.
- **MSc in Mathematics**, Federal University of Bahia (UFBA), 2019–2021.
- **BSc in Mathematics**, Federal University of Minas Gerais (UFMG), 2010–2017.

Publications and preprints

1. D. Erhard, T. Franco, M. Jara and E. Pimenta. "A Functional Central Limit Theorem for the General Brownian Motion on the Half-Line." Accepted, forthcoming in *Bernoulli*. Journal page.

2. J. C. Arroyave, E. Barros and E. Pimenta. "Scaling Limit of Boundary Random Walks: A Martingale Problem Approach." Preprint, 2025. arXiv:2507.10528.

Theses

- **Functional Central Limit Theorem and Berry–Esseen Estimates for Non-Homogeneous Random Walks**. Doctoral thesis, UFBA, 2025. Advisor 1: Tertuliano Franco; Advisor 2: Dirk Erhard.
- **Construction of the Dawson–Watanabe Process**. Master's dissertation, UFBA, 2021. Advisor: Tertuliano Franco.

Selected presentations

- "Scaling Limit of Boundary Random Walks: A Martingale Problem Approach." XVII CLAPEM, 2026.
- "A Functional Central Limit Theorem and Weak Berry–Esseen Estimates for Non-Homogeneous Random Walks." IHES Summer School — Statistical Aspects of Nonlinear Physics, 2025.
- "Scaling Limits of Boundary Random Walks to Feller-Type Processes on the Half-Line." 2025.
- "A Functional CLT for the General Brownian Motion on the Half-Line." 2024.
- "Berry–Esseen Estimates Based on Generators with Applications." 2023.